Understanding Constituent Redistribution, Thermal Transport, and Fission Gas Behavior in U-Zr Annular Fuel Without a Sodium Bond

Technical Work Scope: FL-3
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1 Project Objectives

In this project, we will determine the reason for the distinct constituent redistribution in annular U-Zr fuel without a sodium bond [1,2] and how it changes the fission gas behavior and thermal conductivity. This will be achieved using a combination of microstructure characterization and thermal conductivity measurements of irradiated U-Zr annular fuel and multiscale modeling and simulation using the MARMOT and BISON fuel performance codes. We will use new experimental data and MARMOT simulation results to develop and validate constituent redistribution, fission gas, and thermal conductivity models in BISON that are accurate for both non-annular and annular fuel.

2 Project Scope

Uranium-zirconium (U-Zr) metallic fuels are candidates for Generation IV nuclear reactors and are proposed for use in TerraPower's Natrium^{TM} advanced reactor (selected for the Advanced Reactor Demonstration Program) and the US Department of Energy's (DOE) Versatile Test Reactor (VTR) [3]. Seeing the renewed interest in metallic fuel concepts by vendors, understanding the performance of metallic fuel during reactor operation has increased in priority. These fuels have been extensively investigated in the past but instrument and funding limitations in the 1980s and 90s yielded limited datasets, which are now used to inform new generation models to expedite fuel qualification efforts [4]. The historic tests were typically carried out with a larger gap between the fuel and the cladding than used for LWR fuel, due to the larger swelling in metallic fuel compared to UO_2. To improve the heat transport across the gap, a sodium bond was used. The fuel was designed for a closed fuel cycle in which the sodium bond could be removed, and the fuel recycled. However, an open fuel cycle may be required for new commercial reactor designs.

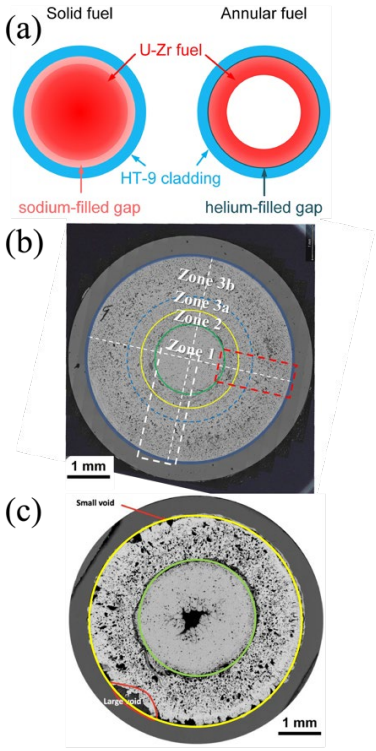


Figure 1. (a) schematic of the solid and annular fuel designs and irradiated slugs of (b) solid U-10Zr [1] and (c) annular U-10Zr [2]. Note the differences in the constituent redistribution and porosity.

In an open cycle, disposability of the fuel is critical, but sodium bonded fuel poses a repository safety risk and increases the disposal cost, since it is mixed waste. Therefore, new fuel designs are being considered with no as-built gap that do not need a sodium bond. Room for swelling is provided by changing the slug geometry, i.e., to an annular design (see Figure 1(a)). However, there is less data on the performance of the new geometries without sodium bond, though fundamental phenomena are generally congruent between these designs. As part of the DOE Advanced Fuels Campaign (AFC), ten annular fuel rods have been irradiated but only one has published characterization data [2,5]. These AFC preliminary data suggest that the constituent redistribution and microstructure evolution in annular fuel may be different than in non-annular fuel slugs with a sodium bond, as shown in Figure 1(b) and (c), though the analyzed rodlet did run at atypically high temperatures. In non-annular fuel, the radial temperature gradient drives Zr redistribution that typically results in three primary zones (Figure 1(b)), with Zr depleted in Zone 2 and enriched in Zone 1. In the annular fuel (Figure 1(c)), material

migrated into the open center region of the fuel, filling most of it. Zr redistribution resulted in only two zones: the outer Zone 2 that was depleted of Zr and the new inner Zone 1 that was enriched with Zr (50 wt% Zr, Figure 1(c)) [2]. BISON simulations of annular fuel that only considered swelling and not constituent redistribution significantly under predicted the shrinking of the center annulus [6]. It is currently unclear why the constituent redistribution is so different in the annular fuel.

The change in the constituent redistribution also changed the fission gas bubble behavior. The size and number density of fission gas bubbles varies in the different fuel zones in non-annular fuel slugs, as shown in Figure 1(b), with the largest bubbles found in Zone 3. This is even more pronounced in the annular fuel, where the bubbles in Zone 2 are 20 times larger than those in Zone 1 [2] (see Figure 1(c)). The change in the bubble behavior directly impacts the fuel thermal conductivity since the bubbles impede heat transport. This decreases our ability to accurately predict the fuel performance of annular fuel slugs because current models of swelling and fission gas release are based on data from non-annular fuel slugs with a sodium bond. They are functions of temperature and fission rate but neglect Zr concentration and phase fractions; thus, they cannot consider the impact of the modified constituent redistribution in annular fuel. In fact, our collaborator TerraPower has determined that the lack of a model of the fuel thermal conductivity that is accurate with and without a sodium bond is a major concern.

The goal of this proposed project is to answer the question of why constituent redistribution in annular U-Zr fuel slugs without a sodium bond differs from that in non-annular fuel with a sodium bond and how it changes the fission gas and thermal conductivity behavior. The annulus was designed to shrink due to swelling, but the fact that a BISON simulation that accounts for swelling underpredicted the shrinking [6] indicates that there are additional mechanisms involved. ***We hypothesize that Zr redistribution and swelling contribute to the shrinking of the center annulus, and that the high porosity in the outer region results in a low thermal conductivity and thus higher fuel temperatures that cause more Zr redistribution and swelling***. We will test this hypothesis using structure and property data from annular fuel and multiscale modeling and simulation using the BISON and MARMOT fuel performance codes. The result will be an improved understanding of the performance of annular metallic fuel without a sodium bond and improved modeling capability in BISON for such geometries.

3 Relevance and Outcomes/Impacts

This project will reveal the cause of the unique constituent redistribution in the AFC annular U-Zr fuel and how it changes the fission gas behavior and thermal conductivity. It will also provide a ***validated and predictive capability in BISON for open-cycle annular U-Zr fuel performance*** for various burnup and temperature conditions. This will include three improved and validated BISON models that will be contributed back to the BISON repository to make them available to all BISON users. PI Tonks has an established record of committing models to BISON [4,7,8]. These three BISON models will be:

- A constituent redistribution model for U-Zr that can include mass transport into a center anulus.
- A mechanistic fission gas release model for U-Zr that couples to the constituent redistribution model and is a function of Zr concentration and phase fractions.
- A thermal conductivity model for U-Zr that couples to the constituent redistribution model and the fission gas model.

These achievements will decrease the time needed to develop and qualify annular fuel designs. They will also overcome current limitations of the U-Zr thermal conductivity and fission models in BISON and improve the accuracy of U-Zr fuel performance simulations for traditional non-annular fuel slugs with a sodium bond. These outcomes will directly address the stated goals in the technical work scope description: “Proposals are sought for metallic fuel modeling interests including …, predictive capability, design streamlining, and licensing support.”

4 Logical Path to Accomplishing Scope

Our goal to reveal the cause of the unique constituent redistribution in annular U-Zr fuel and how it changes the fission gas behavior and thermal conductivity will be accomplished by the completion of three tightly coupled tasks, described in the following sections.

4.1 Task 1: Constituent Redistribution in Annular Fuel Modeling using BISON

BISON currently represents shrinking of the center annulus due to swelling [6], but it under predicts the decrease in the center void shown in Fig. 1(c). We hypothesize that this is because the temperature gradient in the fuel drives mass transport into the center annulus, causing additional shrinking to that caused by swelling. In this task, we will test this hypothesis by adding this additional mechanism of mass transport into our existing CALPHAD-based constitutive redistribution model in BISON [4]. We will then compare its predictions to the characterized final structures in the ten irradiated annular slugs to determine how accurately the hypothesized mechanisms define the true behavior.

4.1.1 Task 1.1 Model development for constituent redistribution into annulus

Our current model uses thermodynamic data from the literature and predicts constitutive redistribution through a traditional fuel slug [4], as shown in Figure 4(a). We assume that the thermodynamic driving forces for the behavior in the annular fuel slug are the same as in non-annular slugs: mass transport via Fickian diffusion and the Soret effect; however, in the annular fuel, mass is transported into previously empty inner region of the annulus, which the current model cannot represent. The annular fuel slug in BISON is discretized with a finite element mesh that can be altered due to deformation but currently no new mesh can be created to account for additional mass transport.

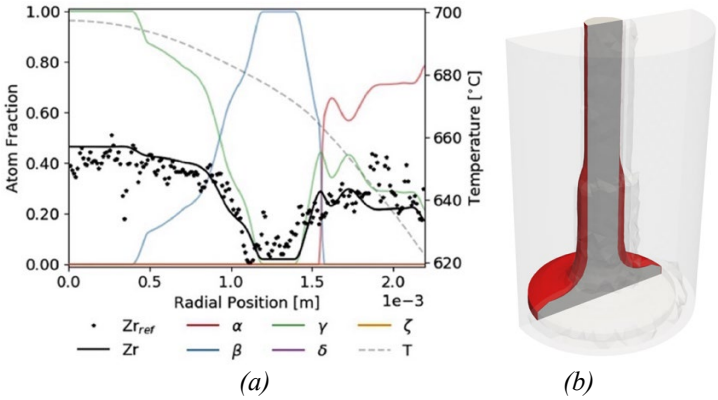


Figure 2: (a) Our current constituent redistribution model in BISON, showing the Zr concentration, phase fractions, and temperature [4]. (b) Growing oxide layer (in red) on an engine valve (gray) using XFEM in MOOSE. Unpublished results of PI Tonks.

We will add the capability to represent mass transport into the empty inner region using the extended finite element method (X-FEM) available in MOOSE, the framework BISON is based on. X-FEM has been used in BISON to model zirconium oxide formation on LWR cladding tubes [9] and oxide formation on stainless steel engine valves in a project led by PI Tonks (see Figure 4(b)). Here, we will use X-FEM to model the migration of the inner fuel surface into the annulus. The use of X-FEM will allow the new solid regions of the fuel to be treated as part of the finite element mesh and so the temperature, displacements, and Zr concentration will all be calculated for the new material. The model will also predict the zirconium redistribution and phase fractions throughout the original and new portions of the annular fuel slug.

4.1.2 Task 1.2 Validation of constituent redistribution model for annular fuel

We will validate the new capability by comparing against the data on the irradiated annular fuel slug available in the literature, from our INL collaborators, and from Task 2. We will directly compare the predicted shrinking of the annulus with the measured value. We will also compare the Zr concentration with that obtained from the experimental characterization. Phase changes occur in the fuel due to cooling as the reactor is shut down, such that the phase fractions in the characterized fuel will differ from what they were during reactor operation. Therefore, we will simulate the cooling of the fuel and the resultant phase transformation and compare the phase fractions in the room temperature fuel with the characterized fractions from transmission electron microscopy (TEM) data already collected by our INL collaborators (see detailed description in Task 2.2). The temperature is the driving force for the constituent redistribution and mass transport into the annulus, and therefore it is impacted by thermal conductivity. Thus, our model predictions will be somewhat inaccurate until the thermal conductivity model is improved in Task 3. Therefore, ensuring that the shrinking of the anulus is more accurate than the previous BISON simulation and quantifying the differences between the predictions and the experimental data will constitute validation at this stage of the work. Validation exercises will be repeated once the thermal conductivity model is complete to ensure that the coupled effects behave as expected and that overall accuracy improves.

How well the new constitutive redistribution model predicts the shrinking of the center annulus compared to the experimental data will indicate the accuracy of our hypothesis that both swelling and mass transport driven by the temperature gradient contribute. The model predictions are impacted by uncertainty in the model parameters and operating conditions, resulting in uncertainty in the model predictions. We will use uncertainty quantification with BISON via the stochastic tools module [10] that is available with BISON. We will also consider the uncertainty in the experimental data when we compare the model predictions. Thus, the validation will be carried out statistically, allowing us to obtain a confidence interval on the predicted values and on the accuracy of our hypothesis. We will ensure that the confidence interval increases when using the new thermal conductivity model from Task 3.

4.2 Task 2: Bubble Structure and Thermal Conductivity Characterization of Irradiated Annular Fuel

Additional analysis of irradiated annular U-Zr fuel slugs without a sodium bond is needed to understand the connection between the constituent redistribution in annular fuel and the fission gas bubbles structures and thermal conductivity. The ten irradiations of annular fuel rods carried out by AFC provide sufficient samples to reveal the connection, but additional analysis of existing data and property characterization is needed. Access to the characterization data and samples from the ten irradiated annular rods will be provided by our INL collaborators. We will use the annular rod data to characterize how the bubble structures change in the annular slugs compared to non-annular slugs and to measure the thermal conductivity in various locations throughout the annular slugs. These data will be used as we develop improved BISON models for the thermal conductivity and fission gas behavior in Task 3.

4.2.1 Task 2.1: Bubble structure characterization

As stated in Section 2, it is clear from the published data for an annular fuel slug [2] that the altered constituent redistribution resulted in changes in the fission gas bubble behavior. In this task we will quantify the changes in the behavior by characterizing the bubble number density and sizes in the various zones in the irradiated fuel in annular fuel slugs and comparing them with similar data from non-annular fuel slugs.

We will leverage the data already collected by our INL collaborators on annular U-10Zr fuel irradiated to ~3% fissions per initial heavy metal atom (FIMA) and non-annular U-10Zr fuel irradiated to 5.7 at. % burnup by AFC and other fuels irradiated as part of the AFC-3A, 3B, 3C, 3D experiments. Specifically, we will utilize the previously collected backscattered electron (BSE) micrographs of fuels collected with a scanning electron microscope (SEM) and optical metallography data, which are standard for baseline postirradiation examination (PIE). These data were collected on several annular U-10Zr fuels.

In this effort, we will use both optical and BSE micrographs to quantify the porosity in each region using automated algorithms that have been previously tested and qualified for use on metallic fuels [5,11,12]. As shown in Figure 3, a BSE micrograph acquired from irradiated metallic fuel can be converted to a binary image needed to segment pores and extract the information about the porosity, equivalent area, equivalent diameter, and the eccentricity of pores and bubbles. We will quantify porosity in each region of the fuels and also utilize the algorithm developed by co-PI Aitkaliyeva's group to investigate the interconnectivity of the pores and estimate the pore size distribution in 3D from 2D images [11,12]. Our primary emphasis will be the annular U-10Zr irradiated to 3%FIMA as it has undergone SEM and TEM examinations and thermal diffusivity measurements as part

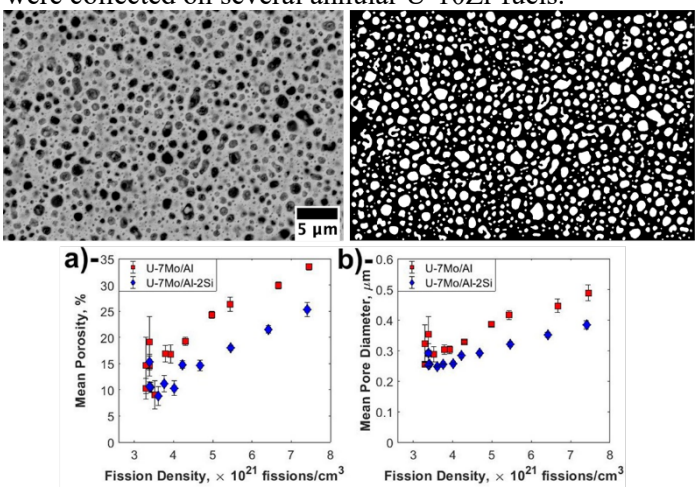


Figure 3: Example of the pore/bubble analysis procedure, where BSE micrograph is converted to binary image that can be used to extract information about porosity. [11,12]

of an AFC characterization campaign. Similar analysis will be carried out on BSE data from non-annular fuel irradiations, to allow for comparison of the bubble structures.

4.2.2 Task 2.2: Thermal conductivity characterization

The thermal conductivity of irradiated U-Zr depends on its temperature, zirconium concentration, the fractions and structure of the phases present, defects, solid fission products, and fission gas bubbles. Thus, the altered constituent redistribution and bubbles structures in the annular fuel will also alter the thermal conductivity. In this task we will measure the thermal conductivity at various radial positions in irradiated annular fuel slugs and compare them with that from non-annular fuel slugs.

Thermal diffusivity measurements have already been collected by our INL co-PIs on the annular U-10Zr fuel irradiated to 3% FIMA and four non-annular U-Zr fuels using the laser-based thermal conductivity microscope (TCM). The TCM is a relatively new technique that provides localized thermal property data at various locations at a resolution of ~50 μm. Figure 4 shows the thermal diffusivity data collected for 3% FIMA annular U-Zr fuel by our INL co-PIs. Our INL collaborators have additional annular fuels that are scheduled for thermal conductivity measurements, and we will leverage these programmatically planned measurements. As part of the proposed effort, we will use the collected diffusivity data to calculate thermal conductivity once porosity quantification has been completed because previously collected conductivities are not reliable due to the large number of pores in the fuel.

As previously mentioned, TEM data for one of the annular U-10Zr fuels is already available to the PIs and microstructural characterization of other fuels is ongoing. If phase identification of the annular U-10Zr fuel requires additional data, we will lift-out specimens for further microstructural characterization in TEM. We will use focused ion beam (FIB) tool to create TEM lift outs and ensure their locations coincide with the areas where TCM measurements will be conducted (e.g., see numbers in white in Figure 4(b)). Co-PIs Aitkaliyeva and Pavlov have successfully utilized this approach in the past to investigate irradiated mixed oxide fuels and have a well-established procedure for the proposed work [13]. FIB and TEM work will be conducted at INL by co-PI Aitkaliyeva, who has a joint appointment at INL, and her students. The phases present will be identified using selective area electron diffraction, which provides crystallographic information and is thus more accurate than chemical compositional analysis. Co-PIs Aitkaliyeva and Yao have utilized this approach on irradiated U-Zr and U-Pu-Zr fuels in the past and are thus well positioned to undertake the task [2,5,14,15]. Students will have summer internships at INL and have exposure to all experimental activities ongoing at INL.

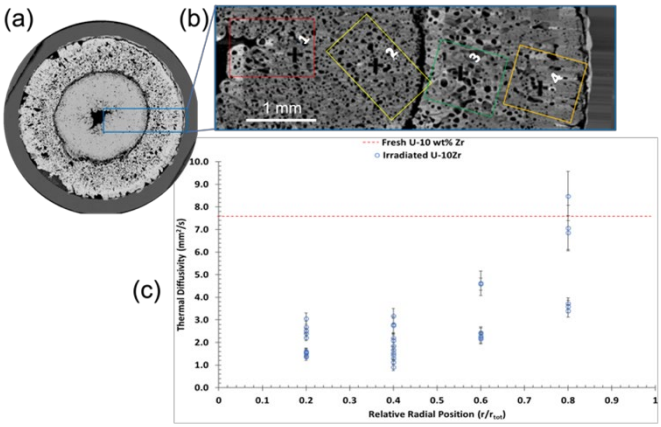


Figure 4: Room temperature thermal conductivity measurements from the annular fuel slug using the TCM, where (a) shows the cross-section of the fuel, (b) shows the regions where TCM data was collected, and (c) shows the measured thermal conductivities as a function of radial position. Unpublished results of co-PI Adkins.

Furthermore, if additional thermal conductivity measurements are required, we will collect data as part of nuclear science user facility (NSUF) experiments. Students will be trained to write NSUF access proposals as lead PIs, which will provide them with a valuable professional development opportunity. Co-PI Aitkaliyeva and her students have successfully used NSUF in the past to gain facility access (obtaining >20 awards since 2017) and conduct experiments at INL. This approach will allow us to directly correlate the thermal properties to the microstructure of the specimen and use both as input to Tasks 1 and 3.

4.3 Task 3: Phase-Dependent Model Development of the U-Zr Thermal Conductivity and Fission Gas Behavior

Fission gas transport and thermal conductivity vary in the different zones that form in U-Zr fuel slugs due to the radial temperature gradient and due to the varying composition and phases present resulting from constituent redistribution. Since the constituent redistribution is distinct in non-annular and annular fuel slugs, the fission gas behavior and thermal conductivity will also be distinct. The current U-Zr models in BISON for fission gas behavior and thermal conductivity do not fully consider these factors and therefore cannot capture the changed behavior in annular slugs. Thus, these models must be improved. We will first assess the performance of the existing models for annular fuel slugs and then modify the mechanistic fission gas and thermal conductivity models for U-Zr.

4.3.1 Task 3.1: Assessment of fission gas and thermal conductivity models for annular fuel slug

Before improving the fission gas and thermal conductivities, their current capacity to represent the behavior of an annular fuel slug must be assessed. These assessments will use the new constitutive redistribution model that will be developed in Task 1.

There are two fission gas models for U-Zr available in BISON. The first is a semi-empirical model [16] and the second is a mechanistic model [17,18]. We will quantify the accuracy of both fission gas models for annular fuel slugs by simulating the behavior of the ten annular fuel rods with BISON and comparing the porosity, bubble radius, and interconnectivity predicted by the model with the porosity and 3D pore size distributions characterized in Task 2. We will also record the computation time of the two models, to compare their computational efficiency.

There are five empirical U-Zr thermal conductivity models available in BISON. The temperature directly impacts the fission gas behavior, and therefore we will repeat the annular fuel assessments using the two fission gas models using each of the thermal conductivity models. We will quantify the resultant error compared to the experimental data from each of the ten separate combinations of fission gas and thermal conductivity models. The assessment results will be used in Tasks 3.2 and 3.3.

4.3.2 Task 3.2: Modification of mechanistic fission gas model

The mechanistic U-Zr fission gas model in BISON considers the concentration of gas in the matrix, diffusion to the bubbles, the hydrostatic stress in the fuel matrix surrounding the gas bubbles, the surface energy of the bubbles, and it uses the van der Waals equation of state for the fission gas within bubbles [17]. Bubble interconnection is defined using a function determined from mesoscale simulations [18]. The U-Zr properties currently used in the model were measured using bulk samples that contained multiple phases and are not functions of the Zr concentration or the fractions of the phases present in the alloy. Therefore, we will modify the mechanistic fission gas model so that it is a function of the Zr concentration and the phase fractions and couples to the constituent redistribution model we will develop in Task 1. We will determine values for the properties as a function of phase fractions using the results from recent literature reporting on atomistic simulations and small-scale experimental techniques that have been applied to determine properties of the individual phases. The surface energy and bulk modulus have been determined as a function of Zr concentration and temperature using atomistic simulations [20,21] and we will use those directly in the fission gas model. Atomistic simulations have also been used to determine the Xe diffusivity for the alpha-U phase [22] and the Xe migration energy for the gamma-U phase [23]. However, the Xe diffusivity through a multiphase structure has not been determined.

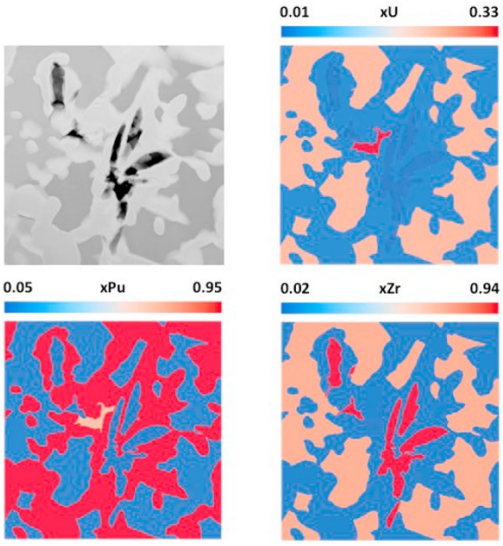


Figure 5: As-cast U-24Pu-15Zr microstructure reconstructed in MARMOT, where the micrograph is shown at the top left and the reconstructed concentrations of U, Pu, and Zr in MARMOT are shown at the top right, bottom left, and bottom right, respectively. [19]

We will determine the effective diffusion coefficients for multiphase microstructures using mesoscale MARMOT simulations that directly represent the multiphase microstructures in various locations of the annular fuel rods. The microstructures will be reconstructed from the irradiated annular fuel that has been characterized by our INL collaborators as part of the AFC program. MARMOT has the capability to directly reconstruct simulation domains from experimental micrographs, as demonstrated in Figure 5 for an as-cast U-24Pu-15Zr microstructure carried out by PI Tonks and Co-PI Aitkaliyeva. We will assign best estimates for the fission gas diffusivity of the phase present at each point throughout the microstructure using values from the literature and then calculate the effective diffusivity across the entire microstructure using existing capability in MARMOT [24]. We will also generate similar multiphase microstructures for other phase fractions not represented in the actual data. We will use the results from the various effective diffusion coefficient simulations to develop an expression for the diffusion coefficient as a function of the phase fraction, Zr concentration, and temperature. The expression will be implemented in the fission gas model.

Once complete, we will redo the assessments from Task 3.1 using the improved mechanistic fission gas model, comparing with the bubble data from Task 2. The new results will be compared with those from the original mechanistic model and from the semi-empirical model to quantify the impact of our modifications. We will also use the new model to simulate existing BISON benchmark cases for non-annular U-Zr fuel slugs and quantify the impact of the modifications.

4.3.3 Task 3.3: Improvement of U-Zr thermal conductivity model and porosity correction

The five empirical U-Zr thermal conductivity models in BISON are fits to legacy data and are functions of temperature and Zr concentration. Recent efforts in the literature have published new thermal conductivity models for U-Zr that have not been implemented in BISON. These include new empirical fits [27], experimental measurements on local regions of the U-Zr microstructure [28,29] and with porosity [30], semi-empirical fits that use data from atomistic simulations [31,32], and mesoscale calculations of effective thermal conductivities of multiphase structures [33]. Thus, there are many potential functions that could be used, and it is unclear which will best represent the thermal conductivity in annular fuel slugs. Therefore, we will compare the thermal conductivity from the five empirical models in BISON and the three new fits to all available U-Zr thermal conductivity data and either select the best model or, if none of the models perform well, create a new semi-empirical model using information from all the various models.

To further evaluate the optimal model, we will also reconstruct mesoscale phase structures in 3D using data from our collaborators and Task 2 and calculate their effective thermal conductivities using existing capability in MARMOT (see Figure 6). We will use individual phase thermal conductivities available in the literature from experiments [28,29] and atomistic modeling [32]. We will compare these effective thermal conductivities from the reconstructed experimental microstructures at room temperature with the corresponding measured values from the TCM (see Figure 4). We will then compare the effective thermal conductivities from our mesoscale model for various temperatures with our selected optimal U-Zr thermal conductivity model to determine if any additional modifications to it are needed. Finally, we will implement the optimal model in BISON and couple it to our new constituent redistribution model from Task 1.

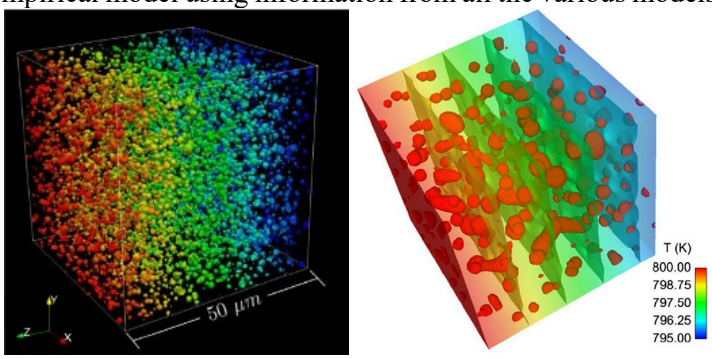


Figure 6: Example of effective thermal conductivity calculations using MARMOT for a composite UO_2 fuel [25] (left) and for a UO_2 structure with intragranular fission gas bubbles [26] (right).

To account for the impact of fission gas bubbles on the thermal conductivity, we will develop an improved thermal conductivity porosity correction by generating 3D bubble structures with various bubble radii, porosities, and interconnections and calculating their effective thermal conductivities with MARMOT. We will also reconstruct 3D experimental bubble structures characterized in Task 2 and

compare their effective thermal conductivities with those from the generated structures. The calculated effective thermal conductivities with porosity will be compared with experimental values from the literature [30]. The calculated effective thermal conductivities with bubbles will be used to generate a porosity correction that couples to the mechanistic fission gas model and uses the local mean bubble radius, porosity, and interconnectivity. We will compare the predictions of our new correction with existing porosity corrections without sodium logging already implemented in BISON. We will then repeat the annular fuel assessment from Task 3.1 using the modified fission gas and thermal conductivity models and comparing with the data from Task 2. We will compare with the initial assessment results to see how the accuracy has improved. We will also repeat the validation of the constituent redistribution model from Task 1.2 to ensure that its accuracy improves. All new models will be committed back to the BISON repository, with the assistance of Co-PI Hirschhorn who is on the BISON team, and will be available to all BISON users.

The improved constituent redistribution, fission gas, and thermal conductivity models in BISON will allow us to investigate how the overall fuel behavior changes in annular fuel slugs. We will carry out a parametric study with the final models to isolate the primary causes of the changed behavior that has been observed [2].

5 Task List and Schedule

Here, we present the work schedule that will be followed in this project. As discussed in Section 4, our project has been broken down into three tasks, and each has several subtasks. The schedule for each task is shown in Figure 7. The tasks and subtasks are:

1. Constituent Redistribution in Annular Fuel Modeling using BISON
   - 1.1. Model development for constituent redistribution into annulus
   - 1.2 Validation of constituent redistribution model for annular fuel
2. Thermal Conductivity and Bubble Structure Characterization of Irradiated Annular Fuel
   - 2.1. Bubble structure characterization
   - 2.2. Thermal conductivity characterization
3. Phase-Dependent Model Development of the U-Zr Thermal Conductivity and Fission Gas Behavior
   - 3.1. Assessment of fission gas and thermal conductivity models for annular fuel slug
   - 3.2. Modification of mechanistic fission gas model
   - 3.3. Improvement of U-Zr thermal conductivity model

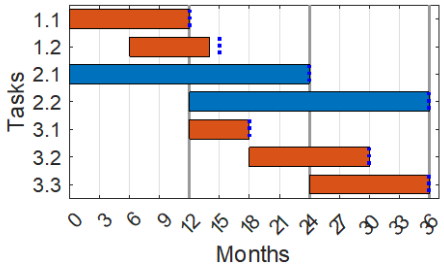


Figure 7: Schedule for the subtasks. Subtasks that will be overseen by PI Tonks are shown in orange and by Co-PI Aitkaliyeva in blue. Milestone due dates are shown with dotted blue lines.

6 Milestones and Deliverables

The major milestones for this project will include:

- Task 1 milestones
  - Constituent redistribution model that includes mass transport into annulus (due Month 12)
  - Error evaluation for new constituent redistribution model by comparing to annular fuel data (due Month 15)
- Task 2 milestones
  - Quantified porosity for irradiated annular and non-annular fuel slugs (due Month 24)
  - Thermal conductivity determination from irradiated annular fuel slug data (due Month 36)
- Task 3 milestones
  - Assessment of current U-Zr fission gas and thermal conductivity models for annular fuel (due end of Month 18)
  - Modified version of the mechanistic BISON U-Zr fission gas model (due Month 30)
  - Finalized U-Zr thermal conductivity model coupled to fission gas and constituent redistribution (due Month 36)

The deliverable for each milestone will be a milestone report submitted with the quarterly report. All new BISON code will be committed to the BISON repository once the corresponding paper is published.

7 Facilities

This project will use various computational and experimental facilities at the various participating institutions. Experimental data from past characterization carried out by AFC will be analyzed using software tools available at UF and new thermal conductivity data will be collected using the TCM at INL. The BISON model development and BISON and MARMOT simulations will be carried out using computational facilities at UF. See the Capabilities document for more information.

8 Roles and Responsibilities

We have assembled a team of researchers for this project with all the expertise required for it to be successful and has a history of collaboration. See the Benefit of Collaboration document for more information. We have researchers from the University of Florida (UF), TerraPower, and Idaho National Laboratory (INL). The overall project will be led by PI Tonks, and he will also lead Tasks 1 and 3. Task 2 will be led by co-PI Aitkaliyeva. Co-PI Hilton from TerraPower will act as the industrial advisor.

In Task 1, the constituent redistribution model will be developed and implemented in BISON by a UF graduate student under the direction of PI Tonks. Co-PI Hirschhorn will help direct the BISON development. Data for validation will be provided from our INL collaborators and from Task 2.

In Task 2, the bubble structures and thermal properties of irradiated annular U-10Zr fuels will be compiled and analyzed. Co-PI Yao will provide optical and SEM data to Aitkaliyeva and another UF graduate student, who will analyze the data and extract information on porosity and phase identification. Co-PIs Pavlov and Adkins will provide measured thermal diffusivity data. Task 2 will provide the provide existing data for the validation for Tasks 1 and 3.

In Task 3, the fission gas model improvements will also be carried out by the UF student under the direction of PI Tonks. The thermal conductivity model improvements will be carried out by a UF postdoc. Hirschhorn will help direct the development of both models. Hilton will work with the UF team to ensure that the new thermal conductivity model meets the needs of TerraPower.

The entire team will meet monthly via video conferencing to present progress, coordinate the modeling and experimental efforts, and address any issues that may arise. The team will also attend the Nuclear Energy Advanced Modeling and Simulation (NEAMS) Program metallic fuel virtual meetings that are held periodically to coordinate with the NEAMS program.

9 Unique Challenges and Planned Mitigations

As challenges arise, they will be addressed by our team, which is very familiar with issues and complications that arise during research. Some of these challenges are likely to require changes to the research plan. If a challenge only impacts the work in a single task, the members of the task team will decide on an appropriate modification. If a challenge impacts the entire project, then the full team will decide on the modification. PI Tonks will make the final decision on changes that affect the entire project, and then present them to DOE in the next quarterly report. Potential challenges include:

- *New constituent redistribution model does not accurately account for shrinking of center annulus*.
  - *Planned Mitigation*: Additional mechanisms for shrinking of the center annulus will be explored. Each will be added to the model and compared against the data. Improvements to the swelling model will also be considered.
- *Delay in obtaining new data on irradiated fuels*. As with any experiments, there could be a delay in obtaining necessary experimental data (e.g., delay in fuel transfer between facilities at INL, instrument operation issue, other unforeseen circumstances).
  - *Planned Mitigation*: In the near term, the necessary data has already been collected and is available for PIs to analyze (e.g., optical metallography, SEM, and TEM results shown in Figs. 1 and 3). Other programmatic work has been planned and scheduled; data will be collected before this project begins, which minimizes the impact on the proposed work. In the worst case scenario, if no new data is collected by the time this project begins, there is still sufficient existing data available to provide data necessary for the validation of the model. The PIs have the experience to address any of the experimental delays and will submit NSUF user facility access proposals to overcome the delays.

- *Mechanistic fission gas model is too unstable to model the annular fuel.*
  - *Planned Mitigation:* The mechanistic fission gas model is more accurate than the semi-empirical model but can also experience numerical difficulties in some conditions. If that occurs for the annular fuel simulations, our team will work with the BISON team to improve its stability. If the stability is still a problem, the semi-empirical model will be used.

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